

Northwind Schema - 4 Database Models

1. PostgreSQL (Relational) - 8 Tables, 8 FK, 69 Fields

```
-- Table 1: categories
CREATE TABLE categories (
    category_id VARCHAR(255) PRIMARY KEY,
    category_name VARCHAR(255),
    description VARCHAR(255)
);

-- Table 2: suppliers (flattened address)
CREATE TABLE suppliers (
    supplier_id VARCHAR(255) PRIMARY KEY,
    company_name VARCHAR(255), contact_name VARCHAR(255),
    contact_title VARCHAR(255), phone VARCHAR(255), fax VARCHAR(255),
    street VARCHAR(255), city VARCHAR(255), region VARCHAR(255),
    postal_code VARCHAR(255), country VARCHAR(255)
);

-- Table 3: shippers
CREATE TABLE shippers (
    shipper_id VARCHAR(255) PRIMARY KEY,
    company_name VARCHAR(255), phone VARCHAR(255)
);

-- Table 4: employees (self-reference: reports_to)
CREATE TABLE employees (
    employee_id VARCHAR(255) PRIMARY KEY,
    last_name VARCHAR(255), first_name VARCHAR(255),
    title VARCHAR(255), birth_date DATE, hire_date DATE,
    phone VARCHAR(255), notes VARCHAR(255),
    street VARCHAR(255), city VARCHAR(255), region VARCHAR(255),
    postal_code VARCHAR(255), country VARCHAR(255),
    reports_to VARCHAR(255) REFERENCES employees(employee_id)
);

-- Table 5: customers (flattened address)
CREATE TABLE customers (
    customer_id VARCHAR(255) PRIMARY KEY,
    company_name VARCHAR(255), contact_name VARCHAR(255),
    contact_title VARCHAR(255), phone VARCHAR(255), fax VARCHAR(255),
    street VARCHAR(255), city VARCHAR(255), region VARCHAR(255),
    postal_code VARCHAR(255), country VARCHAR(255)
);

-- Table 6: products (FK -> suppliers, categories)
CREATE TABLE products (
    product_id VARCHAR(255) PRIMARY KEY,
    product_name VARCHAR(255), unit_price DOUBLE PRECISION,
    units_in_stock INTEGER, discontinued BOOLEAN,
    quantity_per_unit VARCHAR(255),
    supplier_id VARCHAR(255) NOT NULL REFERENCES suppliers(supplier_id),
    category_id VARCHAR(255) NOT NULL REFERENCES categories(category_id)
);

-- Table 7: orders (FK -> customers, employees, shippers)
CREATE TABLE orders (
    order_id VARCHAR(255) PRIMARY KEY,
    order_date DATE, required_date DATE,
    shipped_date DATE, freight DOUBLE PRECISION,
    ship_name VARCHAR(255), ship_address VARCHAR(255),
    ship_city VARCHAR(255), ship_region VARCHAR(255),
    ship_postal_code VARCHAR(255), ship_country VARCHAR(255),
    customer_id VARCHAR(255) NOT NULL REFERENCES customers(customer_id),
    employee_id VARCHAR(255) NOT NULL REFERENCES employees(employee_id),
    shipper_id VARCHAR(255) NOT NULL REFERENCES shippers(shipper_id)
);

-- Table 8: order_details (composite PK, FK -> orders, products)
```

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```
CREATE TABLE order_details (  
    order_id VARCHAR(255) NOT NULL REFERENCES orders(order_id),  
    product_id VARCHAR(255) NOT NULL REFERENCES products(product_id),  
    unit_price DOUBLE PRECISION, quantity INTEGER, discount DOUBLE PRECISION,  
    PRIMARY KEY (order_id, product_id)  
);
```

FK Relations: suppliers->products, categories->products, customers->orders, employees->orders, shippers->orders, orders->order_details, products->order_details, employees->employees (self-ref)

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2. MongoDB (Document) - 1 Root Document 'orders', 4-Level Nesting

Single collection 'orders' with all entities embedded as nested documents/arrays.

```
orders (root document)
|-- _id: string (order_id, primary key)
|-- order_date: date
|-- required_date: date
|-- shipped_date: date
|-- freight: double
|-- ship_name: string
|-- ship_destination: object (Level 1)
|   |-- ship_address: string
|   |-- ship_city: string
|   |-- ship_region: string
|   |-- ship_postal_code: string
|   +-- ship_country: string
|
|-- customer: object (Level 1 - embedded, required)
|   |-- company_name: string
|   |-- contact_name: string
|   |-- contact_title: string
|   |-- phone: string
|   |-- fax: string
|   +-- address: object (Level 2)
|       |-- street: string
|       |-- city: string
|       |-- region: string
|       |-- postal_code: string
|       +-- country: string
|
|-- employee: object (Level 1 - embedded, required)
|   |-- last_name: string
|   |-- first_name: string
|   |-- title: string
|   |-- birth_date: date
|   |-- hire_date: date
|   |-- phone: string
|   |-- notes: string
|   |-- reports_to: string (self-reference)
|   +-- address: object (Level 2)
|       |-- street: string
|       |-- city: string
|       |-- region: string
|       |-- postal_code: string
|       +-- country: string
|
|-- shipper: object (Level 1 - embedded, required)
|   |-- company_name: string
|   +-- phone: string
|
+-- details: array (Level 1 - order line items)
    +-- [each item]:
        |-- unit_price: double
        |-- quantity: int
        |-- discount: double
        +-- product: object (Level 2, required)
            |-- product_name: string
            |-- unit_price: double
            |-- units_in_stock: int
            |-- discontinued: bool
            |-- quantity_per_unit: string
            |-- category: object (Level 3, required)
            |   |-- category_name: string
            |   +-- description: string
            +-- supplier: object (Level 3, required)
                |-- company_name: string
                |-- contact_name: string
```

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```
|-- contact_title: string
|-- phone: string
|-- fax: string
+-- address: object (Level 4 - deepest)
    |-- street: string
    |-- city: string
    |-- region: string
    |-- postal_code: string
    +-- country: string
```

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3. Neo4j (Graph) - 7 Nodes, 7 Relationships, 61 Properties

Nodes (58 node properties):

Node	Key	Properties
categories	category_id (PK)	category_name, description
suppliers	supplier_id (PK)	company_name, contact_name, contact_title, phone, fax, street, city, region, postal_code, country
shippers	shipper_id (PK)	company_name, phone
employees	employee_id (PK)	last_name, first_name, title, birth_date, hire_date, phone, notes, street, city, region, postal_code, country
customers	customer_id (PK)	company_name, contact_name, contact_title, phone, fax, street, city, region, postal_code, country
products	product_id (PK)	product_name, unit_price, units_in_stock, discontinued, quantity_per_unit
orders	order_id (PK)	order_date, required_date, shipped_date, freight, ship_name, ship_address, ship_city, ship_region, ship_postal

Relationships (3 relationship properties):

Relationship	Direction	Properties	Cardinality
SUPPLIES	suppliers -> products	-	Per supplier: 0..n; Per product: 1..1
PART_OF	products -> categories	-	Per product: 1..1; Per category: 0..n
PURCHASED	customers -> orders	-	Per customer: 0..n; Per order: 1..1
SOLD	employees -> orders	-	Per employee: 0..n; Per order: 1..1
SHIPPED_VIA	orders -> shippers	-	Per order: 1..1; Per shipper: 0..n
REPORTS_TO	employees -> employees	-	Per employee: 0..1 (nullable)
CONTAINS	orders -> products	unit_price, quantity, discount	M:N (0..n both sides)

Note: CONTAINS relationship carries order_details properties (unit_price, quantity, discount). REPORTS_TO is a self-referencing relationship on employees. 1..1 cardinality corresponds to NOT NULL FK in PostgreSQL.

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4. Cassandra (Wide-Column) - 8 Tables, Query-Driven Design, 69 Fields

Denormalized, query-driven design. No foreign keys - referenced IDs stored as plain fields. Partition keys determine data distribution; clustering keys determine sort order within partitions.

```
-- Table 1: categories (simple PK)
CREATE TABLE categories (
    category_id TEXT,
    category_name TEXT, description TEXT,
    PRIMARY KEY (category_id)
);

-- Table 2: suppliers (simple PK, flattened address)
CREATE TABLE suppliers (
    supplier_id TEXT,
    company_name TEXT, contact_name TEXT, contact_title TEXT,
    phone TEXT, fax TEXT,
    street TEXT, city TEXT, region TEXT, postal_code TEXT, country TEXT,
    PRIMARY KEY (supplier_id)
);

-- Table 3: shippers (simple PK)
CREATE TABLE shippers (
    shipper_id TEXT,
    company_name TEXT, phone TEXT,
    PRIMARY KEY (shipper_id)
);

-- Table 4: employees (simple PK, reports_to as plain TEXT)
CREATE TABLE employees (
    employee_id TEXT,
    last_name TEXT, first_name TEXT, title TEXT,
    birth_date DATE, hire_date DATE, phone TEXT, notes TEXT,
    street TEXT, city TEXT, region TEXT, postal_code TEXT, country TEXT,
    reports_to TEXT,
    PRIMARY KEY (employee_id)
);

-- Table 5: customers (simple PK, flattened address)
CREATE TABLE customers (
    customer_id TEXT,
    company_name TEXT, contact_name TEXT, contact_title TEXT,
    phone TEXT, fax TEXT,
    street TEXT, city TEXT, region TEXT, postal_code TEXT, country TEXT,
    PRIMARY KEY (customer_id)
);

-- Table 6: products (composite partition: category_id + supplier_id)
CREATE TABLE products (
    category_id TEXT, supplier_id TEXT,
    product_id TEXT,
    product_name TEXT, unit_price DOUBLE, units_in_stock INT,
    discontinued BOOLEAN, quantity_per_unit TEXT,
    PRIMARY KEY ((category_id, supplier_id), product_id)
);

-- Table 7: orders (partition: customer_id, clustering: order_date, order_id)
CREATE TABLE orders (
    customer_id TEXT, order_date DATE, order_id TEXT,
    required_date DATE, shipped_date DATE, freight DOUBLE,
    ship_name TEXT, ship_address TEXT, ship_city TEXT,
    ship_region TEXT, ship_postal_code TEXT, ship_country TEXT,
    employee_id TEXT, -- logically NOT NULL (FK in PG)
    shipper_id TEXT, -- logically NOT NULL (FK in PG)
    PRIMARY KEY ((customer_id), order_date, order_id)
);

-- Table 8: order_details (composite partition: order_id + product_id)
```

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```
CREATE TABLE order_details (  
  order_id TEXT, product_id TEXT,  
  unit_price DOUBLE, quantity INT, discount DOUBLE,  
  PRIMARY KEY ((order_id, product_id))  
);
```

Key Design Summary:

Table	Partition Key	Clustering Key	Query Pattern
categories	category_id	-	Lookup by category
suppliers	supplier_id	-	Lookup by supplier
shippers	shipper_id	-	Lookup by shipper
employees	employee_id	-	Lookup by employee
customers	customer_id	-	Lookup by customer
products	(category_id, supplier_id)	product_id	Products by category+supplier
orders	customer_id	order_date, order_id	Orders by customer+date
order_details	(order_id, product_id)	-	Lookup by order+product